$H_0: \quad p_0(x) = \begin{cases} 1, & x \in (0,1) \\ 0, & x \notin (0,1) \end{cases}$

$H_1: \quad p_1(x) = \begin{cases} \dfrac{e^{1-x}}{e-1}, & x \in (0,1) \\ 0, & x \notin (0,1) \end{cases}$

a) постр. мощный критерий по выборке объема $n=1$:

$G: \{x_{min} < c\}$

$\alpha_1 = P(x_{min} \le c / H_0) = 1 - P(x_1 > c, \dots, x_n > c / H_0)$

$\alpha_1 = 1 - (1-c)^n \le \alpha \Rightarrow (1-c)^n \ge 1-\alpha \Rightarrow c \le 1 - \sqrt[n]{1-\alpha} \Rightarrow \boxed{c = 1-(1-\alpha)^{\frac{1}{n}}}$

$\alpha_2 = P(x_{min} > c / H_1) = 1 - P(x_{min} \le c / H_1) = 1 - (1-(1-P(c))^n) = (1-P(c))^n = \underbrace{\left(1 - \frac{e}{e-1}(1-e^{-c})\right)}^n$

Критерий сост., если при $n \to \infty$ его мощность $W = 1-\alpha_2 \to 1$ (или $\alpha_2 \to 0$).

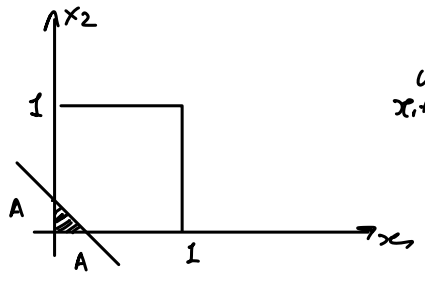
$c = 1-(1-\alpha)^{1/n} \xrightarrow[n\to\infty]{} \frac{\alpha}{n}$

$\Rightarrow \alpha_2 \approx \left(\frac{e^{1-\alpha/n}-1}{e-1}\right)^n \approx \left(\frac{e(1-\alpha/n)-1}{e-1}\right)^n = \left(1 - \frac{e\alpha}{(e-1)\cdot n}\right)^n \to \exp\left(-\frac{e\alpha}{e-1}\right) \ne 0 \Rightarrow$ критерий несост.

b) $l = \frac{L_1}{L_0} = \frac{\left(\frac{e}{e-1}\right)^2 \cdot e^{-x_1} \cdot e^{-x_2}}{1 \cdot 1} \ge c$

$e^{-(x_1+x_2)} \ge B \Rightarrow x_1 + x_2 \le A \quad : G$

$P(x_1 + x_2 \le A / H_0) = \alpha$




$\iint\limits_{x_1+x_2 \le A} dx_1 \cdot dx_2 = \frac{A^2}{2} = \alpha$

$\Rightarrow A = \sqrt{2\alpha}$

$G: x_1 + x_2 \le \sqrt{2\alpha}$

$W = P(x_1 + x_2 \le A / H_1) = \iint\limits_{x_1+x_2 \le A} \left(\frac{e}{e-1}\right)^2 \cdot e^{-x_1-x_2} dx_1 \cdot dx_2 = \dots =$

$= \left(\frac{e}{e-1}\right)^2 \cdot (1 - e^{-A} - A \cdot e^{-A})$

c) Построить наиболее мощный асимптотический критерий:

$l = \frac{L_1}{L_0} = \prod_{i=1}^n \frac{p_1(x_i)}{p_0(x_i)} \ge c$

$\ln l = \sum_{i=1}^n \underbrace{\ln \frac{p_1(x_i)}{p_0(x_i)}}_{\eta_i} \ge \ln c$

$\frac{\sum \eta_i - n \cdot M\eta_i}{\sqrt{n \cdot D\eta_i}} \to N(0,1)$

$P(\ln l \ge \ln c / H_0) = \alpha$

$\eta = \ln\left(\frac{e}{e-1} \cdot e^{-x}\right) = \ln \frac{e}{e-1} - x$

$\ln l = \sum_{i=1}^n \ln \frac{e}{e-1} - \sum_{i=1}^n x_i \ge \ln c \Rightarrow G: \sum x_i \le A$

$P\left(\frac{\sum x_i - n \cdot Mx}{\sqrt{n \cdot Dx}} \le \frac{A - n \cdot Mx}{\sqrt{n \cdot Dx}} \Big| H_0\right) = \alpha$

для норм. распределения: $\frac{A - n \cdot \frac{1}{2}}{\sqrt{n \cdot \frac{1}{12}}} = U_\alpha$

$A = \frac{n}{2} + U_\alpha \cdot \sqrt{\frac{n}{12}}$

$G: \sum_i x_i \le \frac{n}{2} + U_\alpha \cdot \sqrt{\frac{n}{12}}$

$W = P(\sum x_i \le A / H_1) =$

$= P\left(\frac{\sum x_i - n Mx}{\sqrt{n \cdot Dx}} \le \frac{A - n \cdot Mx}{\sqrt{n \cdot Dx}} \Big| H_1\right) =$

$Mx = \int_0^1 x \cdot \frac{e}{e-1} \cdot e^{-x} dx = \frac{e-2}{e-1}$

$Mx^2 = \int_0^1 x^2 \cdot \frac{e}{e-1} \cdot e^{-x} dx = \frac{2e-5}{e-1}$

$Dx = \frac{e^2 - 3e + 1}{(e-1)^2}$

$= \int_{-\infty}^{B} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2}} \cdot dx, \quad B = \frac{\frac{n}{2} + U_\alpha \cdot \sqrt{\frac{n}{2}} - n \cdot \frac{e-2}{e-1}}{\sqrt{n \cdot \frac{e^2 - 3e + 1}{(e-1)^2}}}$

$\alpha_2 = 1 - W$

$B \xrightarrow[n\to\infty]{} \infty$

критерий явл. состоят.

d) $G: x_{min} \le c \qquad H_0: \xi \sim R(0,1)$

$P(\vec{x_n} \in G / H_0) = \alpha$

если $\xi \sim F_0(n)$, то $\xi_{min} \sim 1 - (1 - F_0(n))^n$

$P(x_{min} \le c) = 1 - (1 - F(c))^n = \alpha \Rightarrow \alpha - 1 = -(1-c)^n \Rightarrow c = 1 - \sqrt[n]{1-\alpha}$

$W = P(\vec{x_n} \in G / H_1) = P(x_{min} \le c / H_1) = 1 - (1 - F_1(c))^n =$

$$\left\{ F_1 = \int_0^x \frac{e}{e-1} \cdot e^{-t} \cdot dt = \underbrace{\frac{e}{e-1}(1-e^{-x})} \right\}$$

$$\Rightarrow 1-\left(1-\frac{e}{e-1}\left(1-e^{-1+\sqrt[n]{1-\alpha}}\right)\right)^n = 1-\left(1-\frac{e}{e-1}+\frac{e^{\sqrt[n]{1-\alpha}}}{e}\cdot\frac{e}{e-1}\right)^n \underset{n\to\infty}{\not\to} 1 \Rightarrow$$ критерий не явл. сост.

$$\exp\left(\sqrt[n]{1-\alpha}\right) = \exp\left(e^{\frac{1}{n}\cdot\ln(1-\alpha)}\right) = \exp\left(1+\frac{1}{n}\cdot\ln(1-\alpha)+\bar{o}\left(\frac{1}{n}\right)\right) = e\cdot\left(1+\frac{1}{n}\cdot\ln(1-\alpha)+\bar{o}\left(\frac{1}{n}\right)\right)$$

$$\Rightarrow W = 1-\left(1+\frac{e+\frac{e}{n}\cdot\ln(1-\alpha)+\bar{o}\left(\frac{1}{n}\right)}{e-1}-\frac{e}{e-1}\right)^n = 1-\left(1+\frac{e\cdot\ln(1-\alpha)}{e-1}\cdot\frac{1}{n}+\bar{o}\left(\frac{1}{n}\right)\right)^n \to \exp\left(\frac{e}{e-1}\cdot\ln(1-\alpha)\right) =$$

$$= 1-(1-\alpha)^{\frac{e}{e-1}} \neq 1$$

**T12**

$H_0$: $p_4 = \frac{8}{24}$, $p_3 = \frac{4}{24}$; $p_2 = \frac{6}{24}$; $p_1 = \frac{6}{24}$

$H_1$: $p_1 = p_2 = p_3 = p_4 = \frac{1}{4}$

$\alpha = 0{,}2$
$n = 2$

$$l(\vec{x_n}) = \frac{l_1(\vec{x_n})}{l_0(\vec{x_n})}$$

$H_0$:

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1/16 | 1/16 | 1/24 | 1/12 |
| 2 | 1/16 | 1/16 | 1/24 | 1/12 |
| 3 | 1/24 | 1/24 | 1/36 | 1/18 |
| 4 | 1/12 | 1/12 | 1/18 | 1/9 |

$H_1$:

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1/16 | 1/16 | 1/16 | 1/16 |
| 2 | | | | |
| 3 | | | | |
| 4 | 1/16 | 1/16 | 1/16 | 1/16 |

$G: l \geq c$

$P(l \geq c / H_0) \leq \alpha = 0{,}2$

$\Rightarrow$

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| 1 | 1 | 1 | 3/2 | 3/4 |
| 2 | 1 | 1 | 3/2 | 3/4 |
| 3 | 3/2 | 3/2 | 9/4 | 9/8 |
| 4 | 3/4 | 3/4 | 9/8 | 9/16 |

$p = \frac{1}{36}+\frac{1}{24}+\frac{1}{24}+\frac{1}{24}+\frac{1}{24} = \frac{7}{36} < 0{,}2$

следующий будет перебор $\Rightarrow$ $c = \frac{3}{2}$

$l(1) = 1$; $l(2) = 1$; $l(3) = 1{,}5$; $l(4) = 0{,}75$; Статистика $L(x_1, x_2) = l(x_1)\cdot l(x_2)$

1) $L = 2{,}25$: $\{(3,3)\}$ – $p|_{H_0} = \frac{1}{36}$

2) $L = 1{,}5$: $\{(1,3),(3,1),(2,3),(3,2)\}$ – $p|_{H_0} = 4\cdot\frac{1}{24} = \frac{1}{6} = \frac{6}{36}$

3) $L = 1$: $\{(1,1),(1,2),(2,1),(2,2),(3,4),(4,3)\}$ – $p|_{H_0} = \frac{13}{36}$

$\alpha = 0{,}2$: $\frac{1}{36}+\frac{6}{36} = \frac{7}{36} \approx 0{,}194 < 0{,}2$

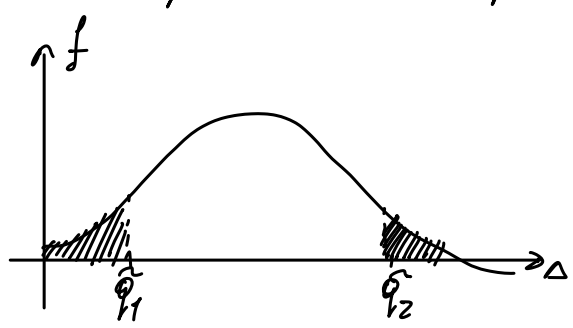
$$G = \left[\begin{array}{l} G_1 = \{(3,3),(1,3),(3,1),(2,3),(3,2)\} \text{ – если встала } \forall \text{ пара – } H_0 \text{ отв.} \\ G_2 = \{(1,1),(1,2),(2,1),(2,2),(3,4),(4,3)\} \text{ – если встала } \forall \text{ пара – } H_0 \text{ отв. с вероятн} \approx 0{,}016 \end{array}\right.$$

**T13**

$\xi \sim N(\theta_1, \theta_2^2)$ ← длина
$\eta \sim N(\theta_1', \theta_2'^2)$ ← ширина

$n = 139$
$m = 1000$

длина: $S_\xi^2 = 5{,}722$; $S_\eta^2 = 6{,}161$ ; ширина: $S_\xi^2 = 4{,}612$, $S_\eta^2 = 5{,}055$




$H_0$: $\sigma_\xi = \sigma_\eta \Rightarrow \frac{S_\xi^2}{S_\eta^2} \sim F(n-1, m-1)$

$H_1$: $\frac{\sigma_\xi}{\sigma_\eta} = \vartheta \neq 1 \Rightarrow \frac{S_\xi^2}{\vartheta^2\cdot S_\eta^2} \sim F(n-1, m-1)$

; тогда $\Delta = \left(\frac{S_\xi}{S_\eta}\right)^2$

$$G: P\left(\left[\begin{array}{l}\Delta \geq \tilde{q}_2 \\ \Delta \leq \tilde{q}_1\end{array}\right| H_0\right) = \alpha$$

$\Rightarrow \Delta \sim a^2 \cdot F(n-1, m-1)$ – ф-ия распр. Фишера

$$W = P\left(\left[\begin{array}{l}\Delta \geq \tilde{q}_2 \\ \Delta \leq \tilde{q}_1\end{array}\right| H_1\right) = \underbrace{\hat{f}\left(\frac{\tilde{q}_1}{a^2}\right)}_{\text{плотность распр. Фишера}} - 1 - \hat{f}\left(\frac{\tilde{q}_2}{a^2}\right)$$

**T14**

$x \sim N(a, \sigma_x^2)$, $y \sim N(b, \sigma_y^2)$, где $\sigma_x^2 = 2$, $\sigma_y^2 = 1$

$H_0: a = b$

$H_1: a > b$

Дисперсии не равны $\Rightarrow$ не можем использовать сл. из распр. Стьюдента.

$\Delta = \bar{x} - a - (\bar{y} - b) \sim N(0, \frac{2}{n} + \frac{1}{m})$

$G: \{\Delta \geq u_{1-\alpha}\}$

По $V$ $\Rightarrow$ $\Delta = \bar{x} - \bar{y} \sim N(0, \frac{2}{n} + \frac{1}{m})$ ; пусть $a - b = \delta$

$\Delta = \bar{x} - \bar{y} - \delta \sim N(0, \frac{2}{n} + \frac{1}{m})$, т.е $\bar{x} - \bar{y} \sim N(\delta, \frac{2}{n} + \frac{1}{m})$

т.е $W(\delta) = P(\Delta \geq u_{1-\alpha})$